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Tangential learning through gamification: A pilot study.

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ABSTRACT

The present study aimed to investigate the possibility of gamification promoting tangential learning and thereby learner autonomy. Hence, with the hypothesis that gamified classes of a topic area would provoke students to do further research in that area outside of class, a pilot study was conducted with an experiment and a control group of 9th grade students from a private school located in Natal/RN, Brazil. Over the course of three weeks, students completed a pretest, underwent pedagogical interventions aimed at acquiring new vocabulary based on sustainability and global warming followed by discussions, after which they completed a post test. Findings showed that students belonging to the Experimental group reported researching about a topic when they find it interesting, at 10% higher rate than the control group. Similar findings were reported with regard to their impressions about learning from games. The study also found that 22% of the experimental group reported researching about the topic out of the classroom more than 3 times, as opposed to 7% of the students in the control group. However, these results, along with classroom observations of student spontaneity suggest that gamification may promote tangential learning by provoking autonomous research behaviour and may be elemental in lowering the affective filter as witnessed in the oral production of the students during the interventions.

Keywords: Gamification. Tangential Learning. Affective Filter. Autonomy. English Use.

INTRODUCTION

As teachers, we want our students to learn as much as they can. However, it is long known, through the advent of Neuroscience, that every student has his or her own agenda; everyone's brain has its own way of working and internalizing new information so as to process new information into knowledge. It is also understood that these processes do not take place only while in the classroom, since learning is an involuntary process that may take place independent of the learner willing to do so or not. Thus, we reach the problem we want to address in this paper: how can learning be promoted out of the classroom?

Portnow and Floyd (2008) have used the term "tangential learning'" to describe the process by which people educate themselves on a topic when it is presented in an enjoyable context (in a game, in a song, in a movie, to name a few). According to these authors, as proposed in their youtube video "Video Games and Learning" (PORTNOW;FLOYD, 2008), when a game introduces content in a way that the player is not "asked" to learn them — the player isn't forced to memorize or process content to continue "playing" — the game just "enables", or facilitates, the learning of the content that the player may ask, without taking the fun away. That poses the main difference between entertainment games to educational games, as the latter aims to teach something while trying to entertain.

We know students can learn tangentially (SOARES, 2014; PORTNOW; FLOYD, 2008), and that it is a fundamental result of student autonomy -- a trait every teacher would love to have developed in their students. Hence, how can we take them in this direction, to learn freely? We believe gamification may be the answer. Gamification, according to Marczewski (2012), has been known and used long before the term was coined by Nick Pelling in 2001. He describes it as making a task more interesting by adopting game mechanics to influence behavior, improve motivation and enhance engagement. Brett Terill (2008) also described the term as "taking game mechanics and applying them to other web properties to increase engagement." According to Huang and Soma (2013), 58% of Americans played video games in 2013. In Brazil, Roberto Gogoni (2015) states that 82% of Brazilians play some game console. Be it video games, cell phones or desktop computers. The secret for the video game's success, is, as it is known, that everything that is fun stimulates the continuity and the progress of the activity in being done. While it seems that students can play

an active role in their educational process and go for knowledge themselves, a question remains: what would motivate them to do that? What may motivate students to spend some of their free time away from any entertainment on behalf of seeking information about something that they are studying at school? Many studies have been carried out in an attempt to answer these questions.

Based on this review and the problems we face due to a lack of student's autonomy in our context of EFL classrooms, we aim to investigate whether using gamification during class can help us promote tangential learning among our students. We hypothesize that the gamified classes may make them We hypothesize that the gamified classes may make them more motivated and engaged in activities, since they would be playing and not become discouraged every time they made a mistake,, but instead feel challenged, thus provoking them to extend this enthusiasm outside of the classroom to further research autonomously on the topics of the gamified classes, thereby promoting tangential learning. This would then provoke them to take their enthusiasm out of the classroom to further research - autonomously - on the topics of the gamified classes, thereby promoting tangential learning.

We want, therefore, to shed some light on what students can tangentially learn from gamified classes, as well as how much English they will feel comfortable using the target language, and, finally, if they would grow any more autonomous out of this experience. Thus, this study aims to answer the following following research questions:

1. What effects of gamification are evidenced in terms of content acquisition in a classroom reading and writing activity in an EFL setting?

2. How does gamification affect spontaneity in using the FL in writing activities?

3. How does gamification promote or not tangential learning?

In order to answer these questions we devised the following research objectives:

1. Identify content acquisition in categories of content knowledge post gamification intervention;

2. Quantify the use of the English pre and post intervention;

3. Verify evidence of autonomous student involvement in gamified class content that may promote tangential learning.

The remainder of this article is organized in the following manner: Section 1 outlines the theories on which this pilot study is based; section 2 describes the methods used to carry

out the research; section 3 presents the results, analysis and discussion of the results, followed by our conclusions, references and annexes.

1. THEORY

Our study is based largely on the concepts of gamification as outlined by Portnow and Floyd (2012). Our study is based largely on the concepts of gamification as outlined by Portnow and Floyd (2012). Gamification, as Portnow and Floyd (2012) define in the video the video "*Gamification*" is simply the idea of taking the principles of play and using them in making real-world activities more engaging, which includes working, studying, getting chores done, any activity. It may be the key to altering the educational system that hasn't changed much since the Prussian model from 1800 since gamification has proven to increase workplace productivity, facilitate learning and even make patients take their medicines on time (PORTNOW; FLOYD, 2012).

Our experience as teachers affiliated to a private school allows us to notice that students who engage into activities and hobbies in the target language (games, music and TV, to name a few) seem to resort on knowledge from such sources for help to fulfil tasks, according to what they say during classes; the opposite seems to be true to students whose contact with English doesn't go beyond the classroom. They are hardly at ease when carrying out classroom tasks and activities, even when it's a review on something that has been already studied or discussed in class.

Aiming to deepen the knowledge on both tangential learning and gamification, this research project seeks to cast some light on (a) how tangential learning can can be promoted in an EFL classroom; (b) what role gamification can can play concerning tangential learning in an EFL class; (c) what students'' logs reveal about tangential learning through gamification.

Hence the term becoming notorious in 2010 in areas such as finance, health, education and productivity as ways to motivate people and increase user activity. "Several vendors now offer 'gamification' as a software service layer of reward and reputation systems with points, badges, levels and leaderboards." (DETERDING et al, 2011).

Hamari et al (2014) provide data to endorse such a perspective in their article "Does Gamification Work? - A Literature Review of Empirical Studies on Gamification", in which they show a graph on the searching hits for gamification, which gives an overview of the

increase of writings on the topic. “It is especially noteworthy that the appearance of the term “gamification” in paper titles has been increasing even more rapidly than general search hits. This suggests that gamification is becoming a more popular subject for academic inquiry.”

In the classroom, gamification can intensify the learning process since the player, during a game, opposite to the reaction people usually have when before an obstacle and frustration moments, feels motivated and challenged (MCGONICAL, 2011). A child, when playing and losing, tends to repeat the phase of that game until he or she reaches success. Now imagine the same happening in the classroom; that means it is possible to promote education with fun and insistence, which would be extremely beneficial to students.

Abrams andand Walsh (2014), in their study, investigated their teenage students’ process of vocabulary development in the subject of English, and reported that their students not only learned with more enthusiasm but they also noticed their own progress, besides developing a certain perception on the language nuances.

Then, the question arises: How would gamification help promote tangential learning? Tangential learning can be defined as exposing someone to new knowledge through which they can instigate learning of their own volition. (PORTNOW, 2008) According to Wilka Soares (2014) such concept considers that part of the audience will self-educate if they are introduced to a theme in an interesting way. Gee (2003) seems to agree as his first revelation from playing video games accounts for games generally requiring the player to learn and think in new ways.

Turkay and Adinolf (2012) point out that while players learn what to do in games, they can also learn knowledge such as physic laws and develop skills such as mathematical thinking, and faster reflexes - indirectly, complementarity, *tangentially*.

> For example, iPirates, a player is given a wind meter which could teach about force vectors. Firing cannons from the side of ships can teach about perpendicular lines. What different types of ships and nationalities were present also communicates to the player in the game. (TURKAY; ADINOLF, 2012)

Prensky (2003) adds that students' attention spans are not short for anything that actually interests them, whilst drawing the difference between Digital Natives (people who are born into the technological world) and Digital Immigrants (people who were born before technology became popular). Digital Natives, according to him, crave interactivity as a result of their experiences. “Traditional schooling provides very little of this compared to the rest of

their world. So it generally isn't that Digital Natives can't pay attention, it's that they choose not to."

As Soares (2014) points out, students' reality has been changing, thus, education must follow that change. It's a researcher's work to evaluate new resources and seek new reasons to utilize them, aiming to promote learning that is adequate to such a new generation of learners.

> The theory of learning in good video games fits better with the modern, high-tech, global world today's children and teenagers live in than do the theories (and practices) of learning that they see in school. Today's world is very different from the world baby boomers like me grew up in and on which we have based many of our theories."(GEE, 2003)

In the results section of her work, Soares (2014) claims that the think-aloud protocols adopted in the research show evidence of tangential learning regarding vocabulary and structures in English. The subjects not only acknowledge but are able to point out new words they learned and strategies they adopted, even failed attempts along the way. Abrams andand Walsh (2014) found similar outcomes in their study - only their research object was gamification. It is then possible that gamification not only helps students learn tangentially, but that it also raises their metalanguage awareness during the process?

As Prensky and Graham (1997) affirm,affirm, "computer games are the quintessential form of experiential and experimental learning. You start out knowing nothing, and gradually, by experimenting and after many errors, you learn to succeed." (PRENSKY;GRAHAM, 1997, p.213). What if we associated *gamification* in order to promote *tangential learning* in the school areas?

This present work is justifiable as we feel there should be more studies regarding gamification, tangential learning and the ELF classroom. Gamification seems to promote effective learning, which occurs while the subjects are happily having fun, making it possible for students to build the same self-confidence that may be built when students decide, on their own, to go deeper into a topic *(tangential learning)*. Both influence the cognitive process as a whole, which makes our investigation extremely relevant.

Gamification has been widely researched nowadays (BORGES, SIMONE DE S., ET AL., 2013; COOK, 2013; FARDO, 2013; KAPP, 2012; REINHARDT AND SYKES, 2014; ROSS, 2015; BORGES, SIMONE DE S., ET AL., 2013; COOK, 2013; FARDO, 2013; KAPP, 2012; REINHARDT AND SYKES, 2014; ROSS, 2015), and its benefits are always

cited as a way to help students control their attention during classes, as a helper to lower affective filters and as a facilitator to reach fluid intelligence.

Tangential Learning, on the other hand, seems to be taking baby steps. Floyd and Portnow (2008) present the idea of what they call tangential learning in games, to which Breuer and Bente (2010) add that digital games, and in particular popular COTS (commercial off-the-shelf) games, can be used to get learners engaged and spark interest for specific topics - which makes us wonder how much can a gamified class do that.

> This notion of 'pull-knowledge' instead of 'push-knowledge' can surely help to foster more self-directed and proactive learning and to help reduce the scepticism of players towards educational/serious games. Consequently, digital games can be learning tools, motivators and generators of curiosity. (BREUER & BENTE, 2010)

2. METHODS

In this section we present a description of the methods that were used to carry out the experiment of this pilot study, including the participant profile, the materials used, the instruments of analysis, and the procedures. The results of the experiment, plus the analysis and discussion follow in section 3.

2.1 2.1 **Participants**

The participants were 51 students from a private school located in Natal, RN, Brazil. They were teenagers,, aging from 14 to 15 years old,, and were enrolled in the 9th Grade. They had curricular English classes, of 50 minutes, five times a week. A group of 28 students served as a control group, which did not engage in gamified learning activities. Another group, of 23 students, comprised the experimental group, which participated in gamified classes.

2.2 Pretests and posttests (Appendix 1)

Participants took pre and posttests,, which entailed entailed reading and answering open-ended questions, before and after the pedagogical intervention. These questions derived from Ogle's, Know/Want to Know/Learned model (1986) in which the students registered what they knew, what they wanted to know and what they learned after the intervention. These questions derived from Ogle's, Know/Want to Know/Learned model (1986) in which the students registered what they knew, what they wanted to know and what they learned after

the intervention. They were also given a questionnaire with a Lickert scale and narrative type questions, where they were asked what they learned about global warming, if they had heard anything about it; they were also asked to explain the meaning of the keywords they studied during the interventions and had two contextualize three of the keywords of their choosing. The data and test scores were calculated and analyzed before feedback was given to each student's parents.

2.3 2.3 Procedures

After receiving the Informed Consent Forms, we proposed to the school board that the tests be applied during class time. They agreed and we carried out the pedagogical intervention described.

The teacher conducted a pretest - which aimed at knowing how much students knew about the topic, how much portuguese words they would use (each word that isn't a preposition or an article would be counted to measure if students had acquire English vocabulary to diminish Portuguese usage), and how well they knew specific keywords related to the topic -, two interventions, and a posttest - whose objective was to measure English language acquisition by comparing not only the quantity of Portuguese words students would use in the posttest, but also how well they would explain the meaning of the keywords and contextualize them.

During the classes, the experimental group was introduced to vocabulary and then played a KAHOOT game - an online quiz where they can score points every time they get an answer right. The control group was introduced to the new vocabulary and was led on a discussion until the moment they were given a worksheet to answer with fill in the blank activities.

Following the next class, the experimental group had a short, gamified, review once again, with one more kahoot session, this time leading to a discussion on the topic, rather than competing for highest scores. The control group received a suggestion of reading a UN article on sustainable tourism by the teacher. This activity was not meant to be scored, and the students knew it.

3. RESULTS AND DISCUSSION

In this section we present the results of the pre and posttests, and a discussion of these

results with regard to finding evidence of tangential learning through gamification.

3.1. KWL Chart and Content Acquisition (Portuguese Usage Included)

The first step taken comprised a pre-test that had three sections. The first section targeted at students' filling in a KWL chart (OGLE, 1986)[1] (OGLE, 1986), in which they were asked to say what they Knew, Wanted to learn and what they actually Learned concerning the topic "Global Warming". The "K" and "W" were measured in the pre-test whereas the "L" was measured by a post-test after students had had a class on the topic. The other two sections of the pre-test aimed at measuring how much the students knew about the meaning of specific keywords and how well they could use them to fill in some gaps.

23 students from the experimental group answered the pre-test KWL questionnaire. For means of measuring English acquisition by the students, they were asked to answer as much as possible in English, but were allowed to use Portuguese whenever they felt like. Then, words in Portuguese used in their texts were counted as well as English keywords, so as to point to their English acquisition progress. In the first section - that asked students (i) what they knew about global warming, and (ii) what they wanted to know about Global Warming - the group used a total amount of 58 Portuguese words to get their message across, and 02 students provided their answers only and fully in Portuguese, their mother tongue. With regard to the other two parts of the pre-test - which aimed at measuring how well they knew the specific keywords that would be taught later on, and if they could fill in gaps with such keywords - the students got an average of 65% of the keywords usage correct. The following is a sample of experimental group student #02 answers in the pretest.

Question 02: Think of global warming. What do you <u>know</u> about the topic? Student #02 from the experimental group answered: The global warming is a problem tô the world. It started happening because of us, the humans, who started using and overusing stuff, as poluentes gases, in industries, cars, etc. This situation affected the nature, of course, and some things started to change, for example, the temperatures started to get higher, the aumento of the nível of the sea, entre outros. **Comment**: The highlights show words in Portuguese used to answer the question.
Question 03: Think of global warming. What do you <u>want to know</u> about the topic? Student #02 from the experimental group answered: I want to know more about it, understand it better, and what is the English expression for "aumento do nível do mar".

[1] The KWL chart is part of the pretest, which is in Appendix I.

Comment: The highlights show words in Portuguese used to answer the question.

Source: our study, Sample of experimental group student #02 answers in the pretest.

In the control group, 28 students used 142 words in Portuguese to get their messages across, and 02 students provided their answers only and fully in Portuguese. With regard to the other two sections of the pretest - which aimed at measuring how well they knew the specific keywords that would be taught later on, and if they could fill in gaps with such keywords - the students from this group got an average of 61% of the keywords usage correct. The following is a sample of control group student #01 answers in the pretest.

Question 02: Think of global warming. What do you <u>know</u> about the topic? Student #01 from the control group answered: - efeito estufa - derretimento das calotas polares - concentração de raios solares **Comment**: The highlights show words in Portuguese used to answer the question.
Question 03: Think of global warming. What do you <u>want to know</u> about the topic? Student #01 from the control group answered: - consequências - como afeta a vida dos seres vivos **Comment**: The highlights show words in Portuguese used to answer the question.

Source: our study; sample of control group student #01 answers in the pretest.

The students were supposed to use keywords in the tests - by keywords we mean the new vocabulary words introduced in the intervention classes. By counting the number of times the keywords were used correctly, we mean to measure their English language acquisition. Hence, after the gamified classes, the 23 students from the experimental group who took part in the experiment, reduced the use of Portuguese words by 80% in the posttest; this means that their use went from 58 Portuguese words in the pretest to a total amount of 12 keywords in the posttest. Some students -as in the following table - who reported wanting to know how to say some vocabulary in English have either researched how to use them in English, or found an alternative way to convey such a message (using antonyms, providing definitions, etc).

Question 01 : Think of global warming. What did you **learn** about the topic?

The global warming is the fenômeno of the temperature of the Earth getting hotter, which makes the geleiras melt, which makes the level of the sea get "bigger", which can cause a lot of consequences, such as tsunamis, ... Also, there are a lot of scientific consequences, in the reefs, in the athmosphere. But I didn't exactly LEARN that in these classes, just refreshed, because I had already heard of it and already knew it.
Comment: Word in Portuguese / Word Learned / Word Substituted / Keyword Used

This sample shows that student #02 used (i) 02 words in Portuguese to answer; (ii) learned one word ("level") that was written in Portuguese in the pretest, (iii) avoided Portuguese (used "bigger" to say "increase"), and (iv) used on of the keywords that were presented in the intervention classes.

Sample of experimental group student #02 answers in the posttest.

Other illustrative examples includes two students[2] who had used only Portuguese in their pretest answered entirely in English on the posttest, clearly showing evidence of English language acquisition.

The control group, which neither had a gamified class nor were asked to research on the topic, also showed a reduction in the use t of the mother tongue[3]. From 142 key vocabulary words used in the pretest, there were only 35 in the posttest, representing a reduction of 76%.

Although there is a higher percentage of English usage in the experimental group it is slight -- 4%. Hence, this indicates the need to do a more systematic and longitudinal study in this context.

3.2. Likert Scale

In the posttest, we wanted not only to evaluate the students' performance regarding Global Warming and the use of English, but we also wanted to know what students views were regarding Gamification and Tangential Learning, for which we have included (i) an open question inquiring if students had seen or heard about Global Warming prior to the posttest, (ii) and seven closed questions in which they used the Likert Scale to answer.

Of the students from the Experimental Group, 69% said that they research about a topic when it interests them, while only 54% of the Control Group either agreed or strongly agreed to that. That is evident within the last two questions in the Lickert Scale questionnaire, when students had to grade how much they agreed to question 5.6: "I researched this topic out

[2] One discourse sample is in the Appendix I section, named "B. Pretest - Experimental Group - Student #07."
[3] One discourse sample is in the Appendix II named "C. Pre-test - Control Group - Student #04."

of class between 1 or 3 times last week". Of the students in the Experimental Group, 26% either agreed or strongly agreed to having researched about the topic a couple of times during the week, while the same was true for only 18% of the students in the Control Group.

Regarding gamification, students were asked if they thought games were a good way to learn about a subject, which turned out to be true for 65% of the students in the Experimental Group, and 54% of the ones in the Control Group. However, only 26% of the Experimental group subjects actually play games for learning, which is lower for the Control Group: only 14% of the students reported playing games for learning.

5. CONCLUSIONS

Our results showed that the Experimental group students used far more English in their answers in posttests after a gamified class intervention than the students in the control group, usually producing longer and more detailed answers. Our results also showed that in spite of not having a gamified class, the students in the control group also increased their use of English from the pretest to the posttest, but to a lesser degree than the experimental group. The experimental group has reported researching more than the control group[4]: while 22% of students in the experimental group researched on the topic, the same is true for only 7% of the students in the control group.

An interesting fact, that makes us think about the power of engagement that a gamified class creates, which regards the students who missed one of the intervention classes in which the key vocabulary was either discussed or taught. Of the students from the experimental group who missed a class[5], 100% (3 out of 3) performed well on the post-test (by 'well' we mean that they could create accurate sentences using the keywords correctly), whereas only 40% (2 out of 5) of the students from the control group who had missed one of the intervention classes presented a similar result[6]. It seems obvious to us that, when the teacher promotes a positive classroom environment and invites students to research on what they know or want to know, this sets a model for learners to go after what they want.

With the production of this paper, we, as teacher researchers, are inclined to conclude that gamification may indeed promote tangential learning. However, the various limitations of this study included that it was a short-term intervention, and future research would be more

[4] Appendix III, section A, graph 5.2; and section B, graph 5.2.
[5] One discourse sample is in the Appendix II named "D. Pre-test - Experimental Group - Student #27."
[6] One discourse sample is in the Appendix II named "E. Pre-test - Control Group - Student #27."

solid with a longitudinal study, offering a longer time period for the intervention. Other limiting variables of this study include comparing each student's history with English, and that would certainly influence our findings. Finally, randomizing the students would have made the results more significant, and we strongly recommend that to be included in the method of a future project. We suggest these topics for future research, as to answer the questions that could not be answered with the present research.

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APPENDIX I

PRETEST

1. O que você conhece sobre aquecimento global?

_____

_____

_____

2. Think of global warming. What do you **know** and **want** to know about the topic?

K - What do you **know**	**W -** What do you **want** to know

3. Match the words with their meaning:

a) Area
b) Atmosphere
c) Feature
d) Glacier
e) Melt
f) Reef
g) Soil
h) Temperature

a) Part of something that stands out.
b) How hot or how cold something is.
c) A series of rocks or coral near the surface of the ocean.
d) It's the air that surrounds Earth.
e) A huge body of moving ice.
f) It's a section of a place often set aside for special use.
g) It's the dirt that covers the ground.
h) It means to change from a solid to liquid.

4. Fill in the blanks with the correct word from the list below:

Area - Atmosphere - Feature - Glacier - Melt - Reef - Soil - Temperature

a) Fences are often used to mark _____ where people are not allowed to go.

b) Airplanes fly through the _____.

c) My sister's eyes are her best _____.

d) Many mountains are formed by moving _____.

e) When the sun is out, the snow begins to _____.

f) This _____ is home to many ocean animals.

g) Most trees and plants need rich, wet _____ to help them grow.

h) The _____ at USA is usually at its highest during summer.

POSTTEST

1. Think of global warming. What did you **learn** about the topic?
2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?
3. Explain in **one sentence** what is the meaning of the following words:

a. Area	b. Atmosphere
c. Feature	d. Glacier
e. Melt	f. Reef
g. Soil	h. Temperature

4. Make **three sentences** with words from the list above:
5. Mark the answer that best suits you about the sentences below:

	Strongly disagree	Disagree	Not sure	Agree	Strongly agree

I only study before exams.	1	2	3	4	5
When I find a topic interesting I research about it.	1	2	3	4	5
I think games are a good way to learn about something.	1	2	3	4	5
I usually play games for learning.	1	2	3	4	5
I only read about a topic when my teacher asks me to.	1	2	3	4	5
I researched this topic out of class between 1 or 3 times last week.	1	2	3	4	5
I researched this topic out of class more than 3 times last week.	1	2	3	4	5

APPENDIX II - Discourse Samples.

A. PRE TEST - EXPERIMENTAL GROUP - STUDENT #02.

1. O que você conhece sobre aquecimento global? Escreva.

The global warming is a problem to the world. It started happening because of us, the humans, who started using and overusing stuff, as poluentes gases, in industries, cars, etc. This situation affected the nature, of course, and some things started to change, for example, the temperatures started to get higher, the aumento of the nível of the sea, entre outros.

2. Think of global warming. What do you know about the topic?

The global warming is a problem tô the world. It started happening because of us, the humans, who started using and overusing stuff, as poluentes gases, in industries, cars, etc. This situation affected the nature, of course, and some things started to change, for example, the temperatures started to get higher, the aumento of the nível of the sea, entre outros.

3. Think of global warming. What do you want to know about the topic?

I want to know more about it, understand it better, and what is the English expression for "aumento do nível do mar".

POST TEST - EXPERIMENTAL GROUP - STUDENT #02.

1. Think of global warming. What did you learn about the topic?

The global warming is the fenômeno of the temperature of the Earth getting hotter, which makes the geleiras melt, which makes the level of the sea get "bigger", which can cause a lot of consequences,
such as tsunamis, ... Also, there are a lot of scientific consequences, in the reefs, in the athmosphere. But I didn't exactly LEARN that in these classes, just refreshed, because I had already heard of it and already knew it.

KEY: Used Portuguese / Vocabulary Learned / Vocabulary Substituted / Keyword used

2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?

Yes. TV, internet... Journalists talking about it, people talking about it, news, points of views, texts...

3. Explain in one sentence what is the meaning of the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. A part of a place
B. The air that surrounds the Earth
C. A characteristic of a person
D. I don't remember
E. To turn from solid to liquid
F. The place where nemo and other fishes live
G. I don't remember, but it's related to ground, plantation..
H. The characteristic of the place/air: cold, hot, warm

4. Make three sentences with the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. I always travel to that area
C. I love her features, such as her wave hair, and her eyes
F. Nemo lives in a reef

5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."

5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

5.1. D	5.2 C	5.3 C	5.4 C	5.5 C	5.6 B	5.7 B

B. PRE TEST - EXPERIMENTAL GROUP - STUDENT #07.

1. O que você conhece sobre aquecimento global? Escreva.

O aquecimento global é causado pela poluição das indústrias, carros, etc. todas essas coisa influênciam para que o aquecimento global lamente. O aquecimentos global é uma causa da poluição.

2. Think of global warming. What do you <u>know</u> about the topic?

-é causado pela poluição dos carros e indústrias
-é uma causa da poluição
-pode ser prevenido com a concientizacao das pessoas e com q falta da poluição

3. Think of global warming. What do you <u>want to know</u> about the topic?

-que as pessoas deixem de incentivar a guerra, pois ela não traz nada bom para as pessoas, por mais que alguém saia vitorioso, ainda assim perdeu algo

POST TEST - EXPERIMENTAL GROUP - STUDENT #07.

1. Think of global warming. What did you learn about the topic?
1. That every time That this happend, all of the parts are prejudicate

2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?
2. Yes, i study this with my teacher walcley

3. Explain in one sentence what is the meaning of the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A: a part of sometging
B: where the ar abd the gass stay
C: afther the present
D: the cold part
E: i dont know
F: dont know too
G: What is use to use whith combustível

H: ir ia cole or hot

4. Make three sentences with the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

The cars use soil
The ar stay um atmosphere
I live in a big area

5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."
5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

5.1. B	5.2 C	5.3 E	5.4 B	5.5 C	5.6 C	5.7 C

C. PRE TEST - CONTROL GROUP - STUDENT #04.

1. O que você conhece sobre aquecimento global? Escreva.

O aquecimento global vem acontecendo durante o desenvolvimento mundial. Consequentemente, com a liberação de gases poluentes e resíduos sólidos urbanos descartados indevidamente so fazem aumentar o risco do descongelamento da geleiras e o risco de contaminação de doenças por conta da poluição pelo Ar.

2. Think of global warming. What do you <u>know</u> about the topic?

• help in descongelamento das geleiras.
• the level of the oceans will get mais alto que o normal, aumentando o risco de enchentes
• the temperature global will get worse

3. Think of global warming. What do you <u>want to know</u> about the topic?

• how to get the global warming a little bit invasivo que atualmente
• how to chance the mentally of the people that administram empresas that the global warming isn't just another problem, that is a enormous trouble that will atrapalhar the life of the futures gerações.

POST TEST - CONTROL GROUP - STUDENT #04.

1. Think of global warming. What did you learn about the topic?
I learned that the global warming is more next than never. And if we don't do something rusgar now, we gonna suffer as consequências in the future.

2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?

I didn't hear

3. Explain in one sentence what is the meaning of the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. It's a determined place
B it's the air of round the earth
C. It's what define yourself
D. It's a peace of ice in the ocean
E. It's when the ice cream get liquid
F. It's what exist in Recife
G. Like vinagre
H. When you are sick your temperature get Alta

4. Make three sentences with the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. Don't get there it's a restrict area
B. The temperature of the earth it's getting worse every year
C. As geleiras of the ocean artic its melt

5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."
5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

5.1. E	5.2 D	5.3 E	5.4 D	5.5 D	5.6 B	5.7 A

STUDENTS WHO MISSED ONE OF THE INTERVENTION CLASSES

D. POST TEST - EXPERIMENTAL GROUP - STUDENT #27.

1. Think of global warming. What did you learn about the topic?

I missed this class

2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?

No answer.

3. Explain in one sentence what is the meaning of the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. Delimited place
B. Air in the earth
C. Characteristic
D. Huge body of moving ice
E. Solid to liquid
F. Rocks in the sea
G. Dirt that covers the ground
H. Hot and cold

4. Make three sentences with the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

H. The temperature is hot
B. We are in the atmosphere
A. The area its big

5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."
5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

5.1. D	5.2 C	5.3 E	5.4 B	5.5 D	5.6 B	5.7 B

STUDENT WHO MISSED ONE OF THE INTERVENTION CLASSES

E. POST TEST - CONTROL GROUP - STUDENT #27.

1. Think of global warming. What did you learn about the topic?
Nothing I faltei this class

2. Did you hear about global warming out of classroom? Where and how (TV, internet, magazines, etc.)?
Yes, in television

3.
A. Is the specific local
B. Atmosphere is the camada de gas oxigenio
C.
D. Is the local very cold
E.
F.
G. Is the local when a gente pisa
H. Temperature is the Grau que está certo local

4. Make three sentences with the following words:

A. Area	B. Atmosphere	C. Feature	D. Glacier
E. Melt	F. Reef	G. Soil	H. Temperature

A. In my house have differents area
H. Natal are temperature agradável
B. When vivemos na atmosphere

5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."
5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

5.1. D	5.2 D	5.3 D	5.4 D	5.5 B	5.6 B	5.7 A

APPENDIX III

A. Experimental Group - POST TEST LICKERT SCALE RESULTS (Experiment Group = 23 subjects)

Question 5. Write the answer that best suits you about the sentences below:

a) Strongly disagree	b) Disagree	c) Not sure	d) Agree	e) Strongly Agree

5.1. "I only study before exams."
5.2. "When I find a topic interesting I research about it."
5.3. "I think games are a good way to learn about something."
5.4. "I usually play games for learning."
5.5. "I only read about a topic when my teacher asks me to."
5.6. "I researched this topic out of class between 1 or 3 times last week."
5.7. "I researched this topic out of class more than 3 times last week."

Figure 5.1
Experiment Group. Total of 23 students.
65% (15 out of 23 students) either agree or strongly agree to only studying before exams.

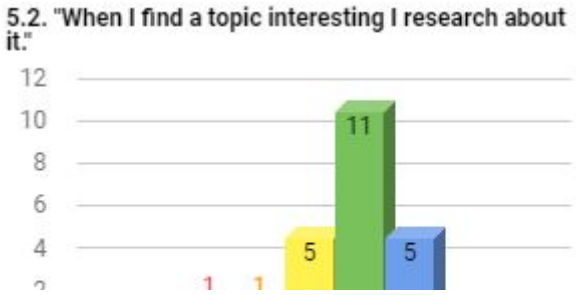

Figure 5.2
Experiment Group. Total of 23 students.
69% (16 out of 23 students) either agree or strongly agree to researching when finding a topic interesting.

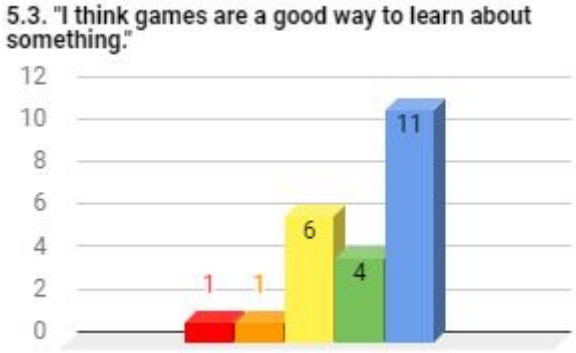

Figure 5.3
Experiment Group. Total of 23 students.
65% (15 out of 23 students) either agree or strongly agree that games are a good way to learn about something.

Figure 5.4
Experiment Group. Total of 23 students.
26% (6 out of 23 students) either agree or strongly agree to usually playing games for learning, whereas 48% (11 out of 23) disagree to that.

Figure 5.5
Experiment Group. Total of 23 students.
22% (5 out of 23 students) either agree or strongly agree to only reading about a topic when a teacher asks them to.
43% (10 out of 23 students) either disagree or strongly disagree to only reading about a topic when a teacher asks them to, whereas 35% (8 out of 23 students) aren't sure.

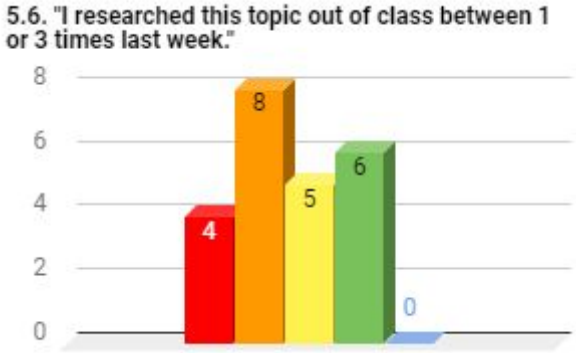

Figure 5.6
Experiment Group. Total of 23 students.
26% (6 out of 23 students) agree to having researched the topic a couple of times during the previous week.
Most of the students, 52% (12 out of 23) either disagree or strongly disagree to having researched the topic a couple of times during the previous week.

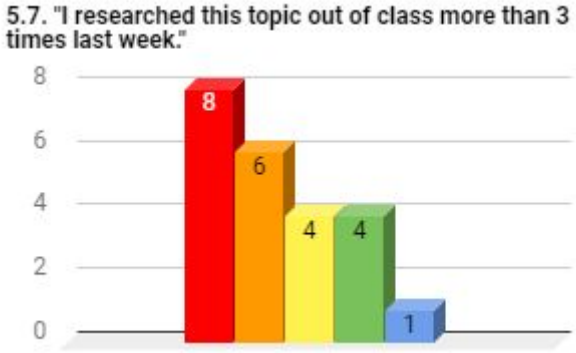

Figure 5.7
Experiment Group. Total of 23 students.
22% (5 out of 23 students) agree to having researched the topic more than 3 times during the previous week.
Most of the students, 60% (14 out of 23) either disagree or strongly disagree to having researched the topic a couple of times during the previous week.

B. Experimental Group - POSTTEST LICKERT SCALE RESULTS (Control Group = 28 subjects)

Figure 5.1
Control Group. Total of 28 students.
43% (12 out of 28 students) either agree or strongly agree to only studying before exams.

Figure 5.2
Control Group. Total of 28 students
54% (15 out of 28 students) either agree or strongly agree to researching when finding a topic interesting.

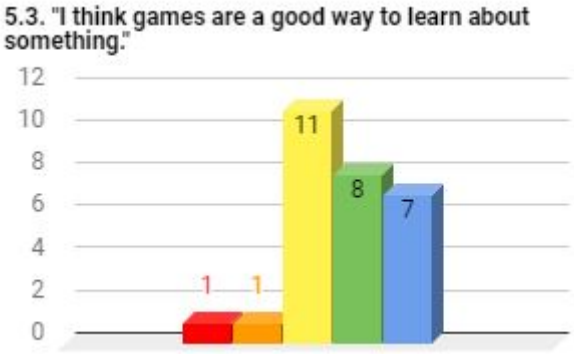

Figure 5.3
Control Group. Total of 28 students
54% (15 out of 28 students) either agree or strongly agree that games are a good way to learn about something.
40% are unsure whether games are a good way to learn about something.

Figure 5.4
Control Group. Total of 28 students
14% (4 out of 28 students) either agree or strongly agree to usually playing games for learning, whereas 60% (17 out of 28) disagree to that.

Figure 5.5
Control Group. Total of 28 students
18% (5 out of 28 students) either agree or strongly agree to only reading about a topic when a teacher asks them to.
61% (17 out of 28 students) either disagree or strongly disagree to only reading about a topic when a teacher asks them to, whereas 28% (8 out of 28 students) aren't sure.

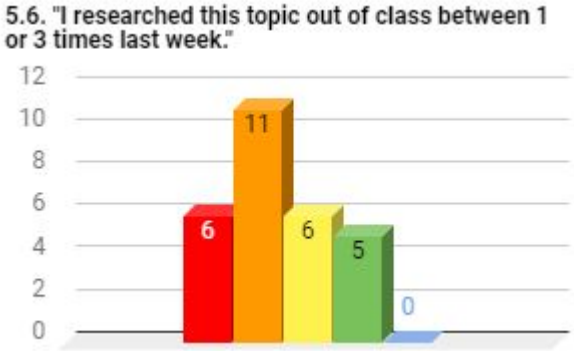

Figure 5.6
Control Group. Total of 28 students
18% (5 out of 28 students) agree to having researched the topic a couple of times during the previous week.
Most of the students, 61% (17 out of 28) either disagree or strongly disagree to having researched the topic a couple of times during the previous week.

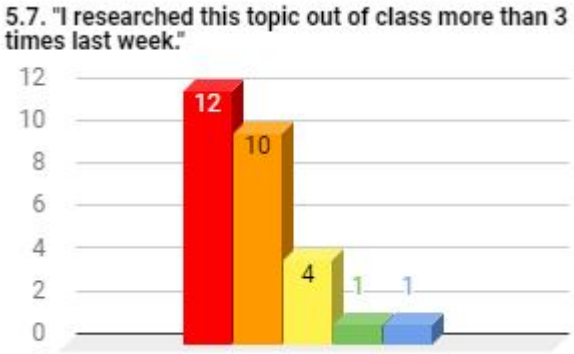

Figure 5.7
Control Group. Total of 28 students
7% (2 out of 28 students) agree to having researched the topic more than 3 times during the previous week.
Most of the students, 95% (22 out of 28) either disagree or strongly disagree to having researched the topic more than three times during the previous week.

DATA COMPARISON CHART

Question	Experimental Group Answers	Control Group Answers
5.2	69% researches	54% researches
5.3	65% believe they learn from games	54% believe they learn from games
5.4	48% don't play games to learn	61% don't play games to learn
5.5	43% read regardless of teacher asking	61% read regardless of teacher asking
5.6	26% read about the topic 1 to 3 times	18% read about the topic 1 to 3 times
5.7	22% had read more than 3 times	7% had read more than 3 times